Athul Shaji

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OBJECTIVE

To pursue an industry or research position focused on computer-based modeling and simulation of physical phenomena, leveraging strong analytical skills, numerical methods, and electromagnetic modeling experience to solve complex engineering problems, while continuously expanding expertise in advanced computational techniques and multidisciplinary research environments.

TECHNICAL FOCUS

Computational Electromagnetics · EM–Circuit Co-Simulation · Nonlinear Time-Domain Analysis · Power Integrity · Numerical Solver Development

EDUCATION

Doctor of Philosophy (PhD), Computational Electromagnetics
Deparment of Electrical Communication Engineering, IISc Bangalore — January 2020 – present

- **Research Summary** - My research lies at the intersection of numerical analysis and computer-aided design, with a focus on time-domain computational methodologies for analyzing nonlinear effects in electronic circuits under incident electromagnetic fields. As part of my PhD work, I developed a complete in-house MATLAB framework to model and simulate these coupled electromagnetic–circuit interactions. Commercial simulation and design tools, including Keysight ADS and CST Microwave Studio, are systematically employed to validate and benchmark the numerical results obtained from the developed methodology, ensuring accuracy, robustness, and practical relevance.
- **Advised by Prof. K. J. Vinoy**
- CGPA: 8.00 (out of 10)

Master of Technology, Electronics
Deparment of Electronics, Cochin University of Science & Technology — April 2009

- Graduatd with First class with Distinction
- Completed a project titled *Blind Signal Separation Algorithms*, as part of thesis work.
- Presented a seminar talk on *Wide-band Leaky Wave Antennas*, as part of the program.
- IEEE Student Branch Secretary of CUSAT

Bachelor of Technology, Electronics and Communication Engineering
T.K.M.Institute of Technology, affiliated to Cochin University of Science & Technology — June 2006

- First Class
- Successfully completed the project on *Fault Detection and Annunciation for Marine EM Log System*, as part of the program.
- Presented a seminar on *Holographic Data Storage*, as part of the program.
- Graduated with First Class.

PROFESSIONAL EXPERIENCE

Project Associate, *Indian Institute of Science*, Bangalore, India — June 2019 – December 2019

- From June 2019 I am working as a project associate in the Microwave Lab, ECE department, IISc Bangalore. The lab is headed by Prof. (Dr.) K. J. Vinoy (https://www.linkedin.com/in/kj-vinoy-a9013339/). Developed an in-house 3D FDTD solver for EMI analysis as part of a DRDO-sponsored project on high-power microwave effects, enabling time-domain evaluation of radiative coupling and circuit-level non-linearity studies. This forms part of the joint project between IISc and DRDO titled "Coupled Electromagnetic Device Circuit Modeling for High Power Microwave Effects".

Industrial Engineer, *Radical Innovations group – RIG*, Finland — January 2019 – May 2019

- *pro bono* work
- vendor interaction

Assistant Professor, *Amrita Vishwa Vidyapeetham*, Kerala, India — July 2015 – December 2018

- Taught both theory and laboratory courses to UG students
- Member of the technical committee to evaluate UG and PG projects

Project Associate, *Indian Institute of Science* Bangalore, India — September 2014 – June 2015

- Developed a C++ executable for performing large scale DC power integrity analysis of PCBs. This DC power integrity tool was developed for *Altium*.
- This PI for this work was **Prof Dipanjan Gope**, Associate Professor, ECE, IISc Bangalore.

Assistant Professor, *Amrita Vishwa Vidyapeetham*, Kerala, India — July 2012 – September 2014

- Taught both theory and laboratory courses to UG and PG students
- Member of the technical committee to evaluate UG and PG projects
- Actively supported the department in the documentation process for NAAC accreditation process

Assistant Professor, *CAPE*, Kerala, India — September 2009 – December 2011

- Member of the technical committee to evaluate UG projects
- Actively supported the department in the documentation process for *TEQIP (Phase-II)*

MOOC

Computational Electromagnetics and Applications (CEMA), *NPTEL* — July 2018 – October 2018

- Topper in the course
- Developed a 2D - FEM based code for solving the Laplace equation on three different geometries
- Actively participated in the discussion forum for the course

SKILLS

- **MATLAB** (good working knowledge) - Developed an in-house MATLAB-based computational framework integrating **FDTD electromagnetics with SPICE-level circuit modeling** to perform time-domain analysis of coupled EM–circuit interactions under external electromagnetic radiation, enabling accurate investigation of radiation-induced nonlinear effects in electronic systems.
- **C++** (good working knowledge) - Developed a **C++-based DC power integrity solver for Altium PCB designs**, incorporating **parallelization techniques** to significantly accelerate execution and enable efficient analysis of large-scale power distribution networks.

- **CST Microwave Studio** (good working knowledge) - Hands-on experience with **CST Microwave Studio** for the **3D electromagnetic modeling, simulation, and design of RF/microwave circuits and devices**, including time- and frequency-domain analysis and result-driven design optimization.
- Keysight ADS - Hands-on experience with **Keysight ADS** for the **modeling and design of RF/microwave circuits and devices**, including advanced **nonlinear analysis using Harmonic Balance and Large-Signal S-Parameter (LSSP) techniques** for accurate characterization of circuit behavior under large-signal excitation.

ACHIEVEMENTS

- **UGC – NET for Lectureship**: *September 2012*
- **GATE**: *2009, 2007*

PUBLICATIONS

International Conferences

1. **A. Shaji** and K. J. Vinoy, "Coupled EM-Circuit Domain Analysis of Radiatively Excited Non-Linear Circuit" 2025, presented at MAPCON 2025.
2. **A. Shaji** and K. J. Vinoy, "Single-Shot Method for the Analysis of Non-Linear Circuits in the Time Domain," 2023 IEEE International Symposium On Antennas And Propagation (ISAP), Kuala Lumpur, Malaysia, 2023, pp. 1-2, doi: 10.1109/ISAP57493.2023.10389003.
3. **A. Shaji** and K. Sankaran, "Thermal integrity modelling using finite-element, finite-volume, and algebraic topological methods," AIP Conference Proceedngs, 2019
4. P. Shandilya, **A. Shaji** and K. Sankaran, "Nanoscale engineered surfaces for cooling in electronic devices," AIP Conference Proceedngs, 2019
5. **A. Shaji** and S. K. Menon, "Artificial Anisotropic Material for Gain Enhancement of Vivaldi Antenna," 2018 Second International Conference on Advances in Electronics, Computers and Communications (ICAECC), Bangalore, 2018, pp. 1-6.
6. **A. Shaji** and R. Kumar, "Joint Polarimetric-Three Dimensional Imaging Using Spiral Phase Mask and Microgrid Polarimeter," 2018 Second International Conference on Advances in Electronics, Computers and Communications (ICAECC), Bangalore, 2018, pp. 1-6.
7. Shilpa K. J. and **A. Shaji**, "A Study on the Accuracy of the Parameters Estimated from a Joint Three-Dimensional Polarimetric Image in the Presence of Poisson and Speckle Noise," 2018 3rd International Conference on Communication and Electronics Systems (ICCES), Coimbatore, India, 2018, pp. 371-375.
8. M. L. Liya, **A. Shaji**, V. M. Jayakrishnan and A. Nair, "Petal shaped semicircular resonator for RFID tags," 2017 IEEE International Conference on Smart Technologies and Management for Computing, Communication, Controls, Energy and Materials (ICSTM), Chennai, 2017, pp. 313-316.
9. Bharath S., Namitha K., M. A. Bharadwaj, **A. Shaji**, "Design of a Zero Index Material Unit Cell and a Study on Its Role in the Gain Enhancement of Vivaldi Antenna," IEEE Conference, 2018, yet to be published.

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